



Introduction to Deep Learning

“Unlocking the Power of Artificial Intelligence”

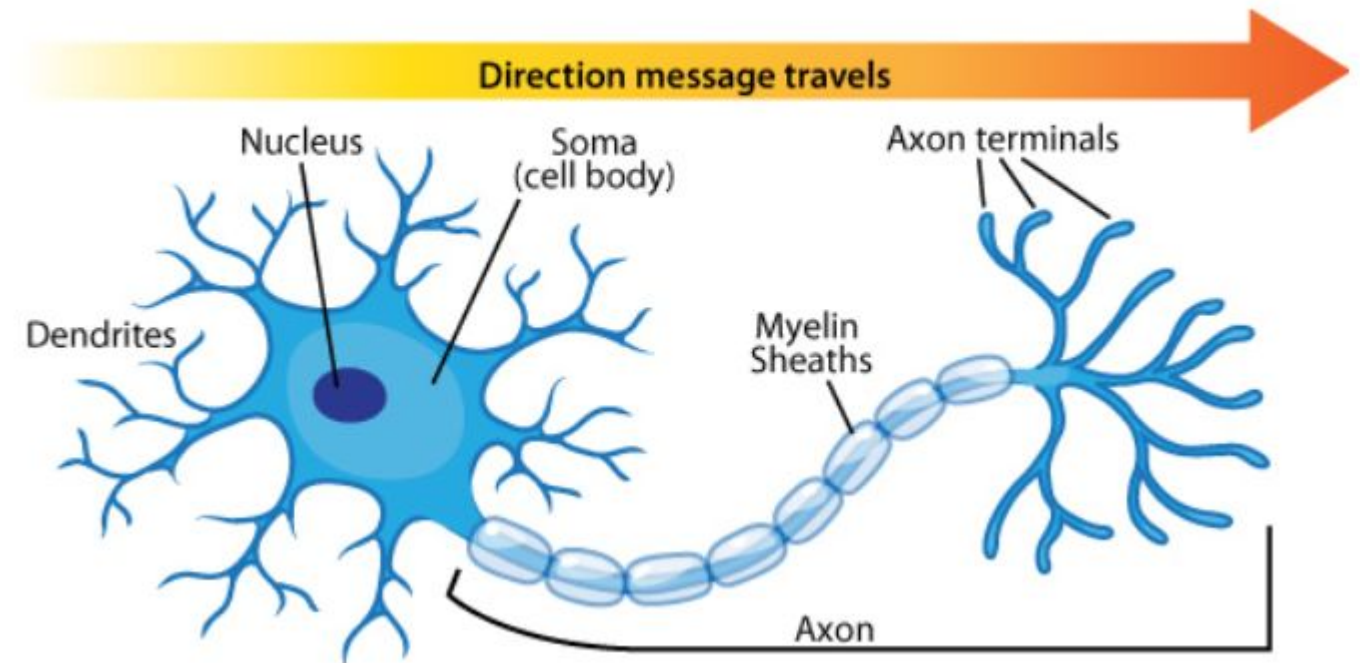
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1. Introduction to Neural Network

Understanding Neurons

- Neurons are the basic units composing the nervous system and are one of the most essential parts of the brain.
- A neuron has the following three basic parts:
 - Cell body - Soma
 - Cell extension - Axon
 - Cell extension - Dendrite
- Neural networks work based on the mechanism of neurons!



1. Introduction to Neural Network

Sample problem

“Assume that you need to build a model to make a decision on whether you should go out for a drink tonight.”

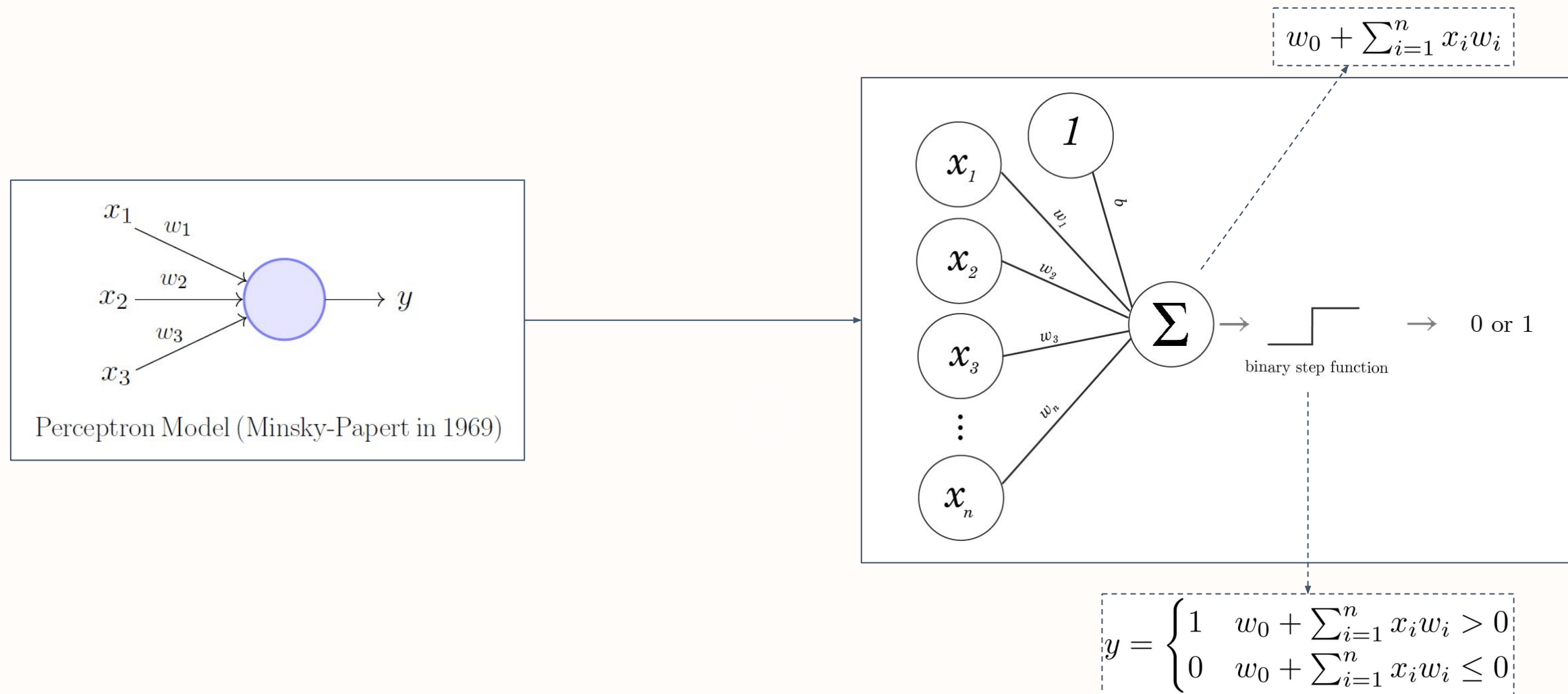
Input feature:

- How much money will it cost?
- Will it consume too much time?
- Is the restaurant far from your location?
- Do you have a deadline tonight?
- Will there be any women in the party?
- Is your wife/GF happy?
- ...

1. Introduction to Neural Network

Basic concepts in NN

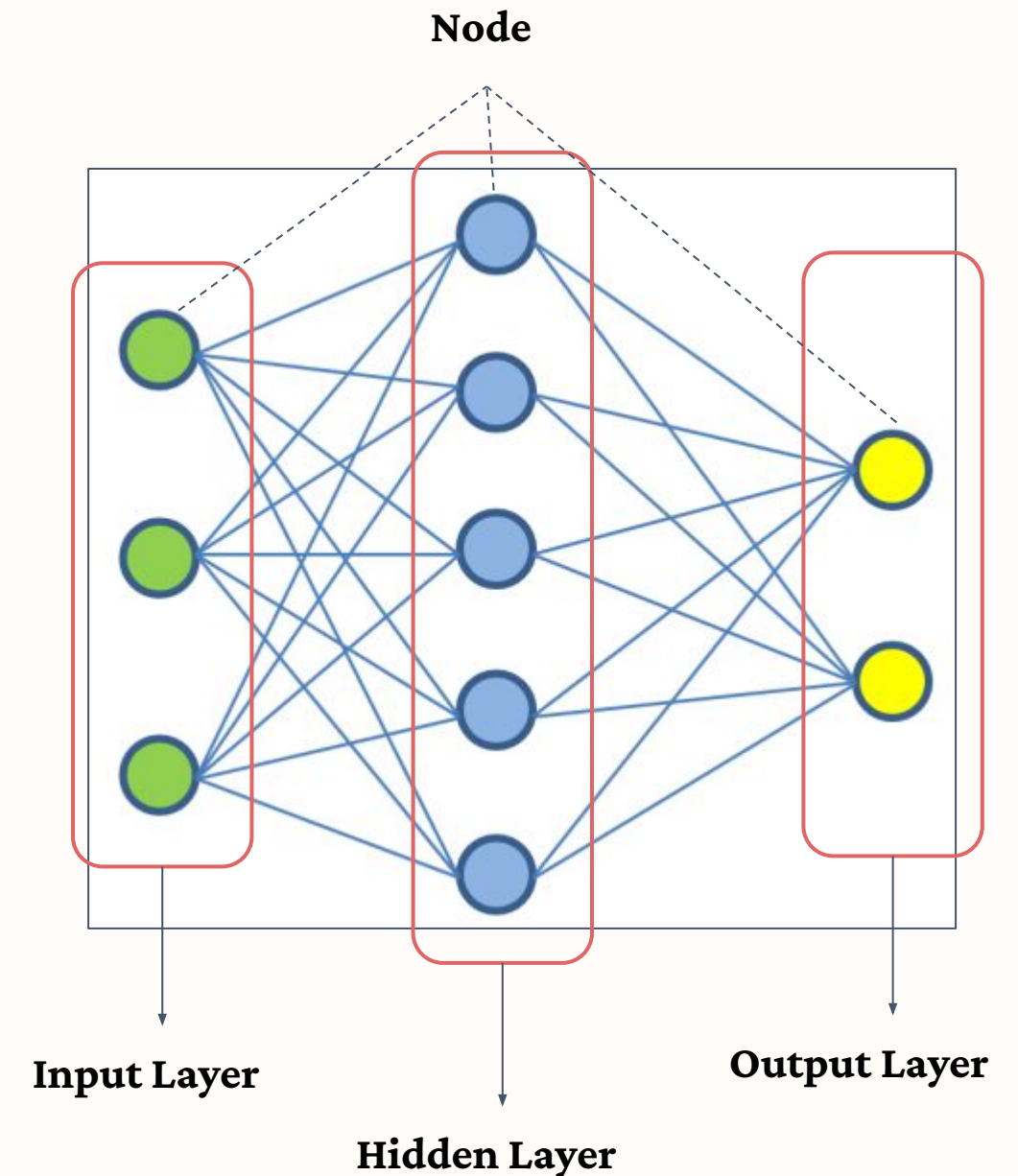
- **Perceptron** is one of the simplest neural network architectures.



1. Introduction to Neural Network

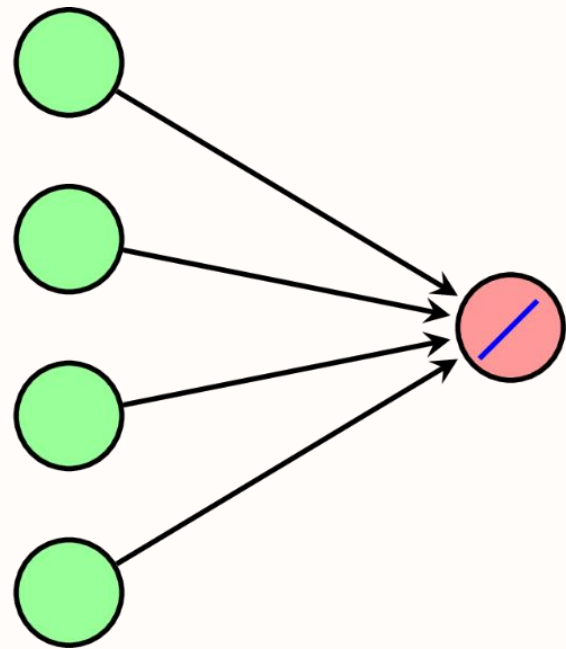
Basic concepts in NN

- A **multi-layer perceptron** is called **Neural Networks (NN)**.
- A basic neural network has interconnected artificial neurons in three layers:
 - *Input Layer*: processes the data, analyze or categorize it, and pass it on to the next layer.
 - *Hidden Layer*: analyzes the output from the previous layer, processes it further, and passes it on to the next layer.
 - *Output Layer*: gives the final result of all the data processing by the artificial neural network

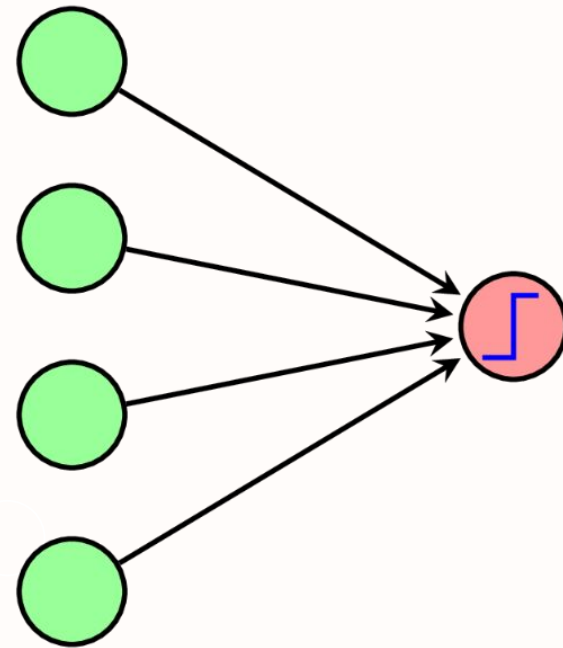


1. Introduction to Neural Network

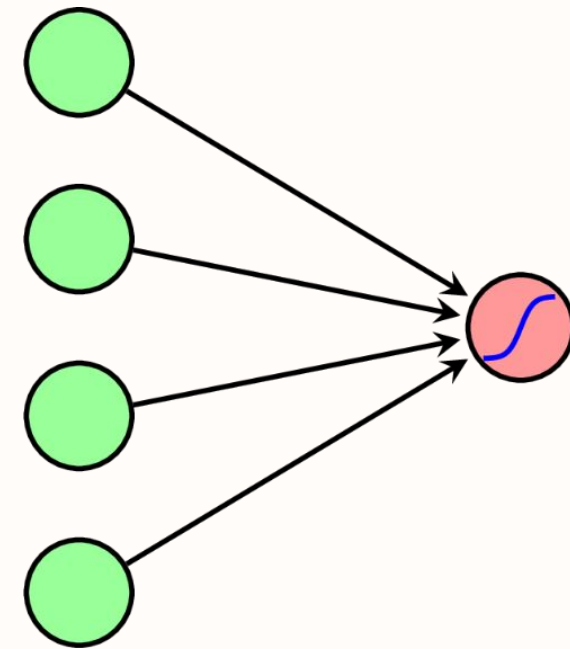
Neural Network in Machine Learning



Linear Regression



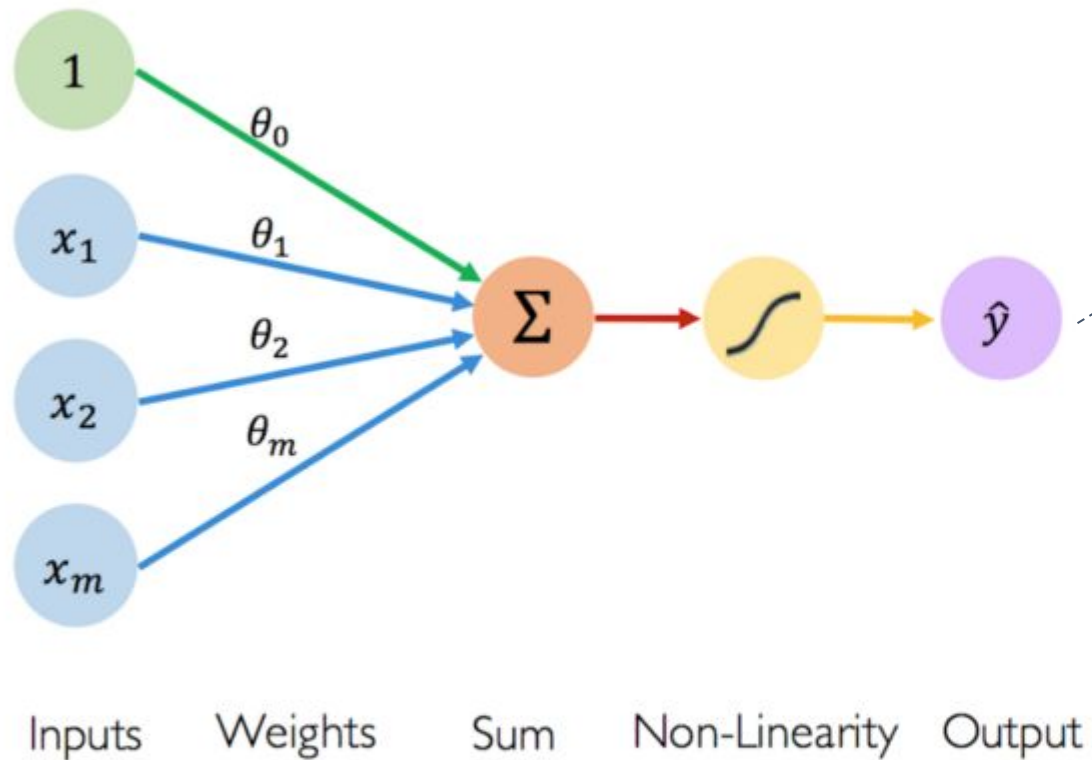
PLA



Logistic Regression

1. Introduction to Neural Network

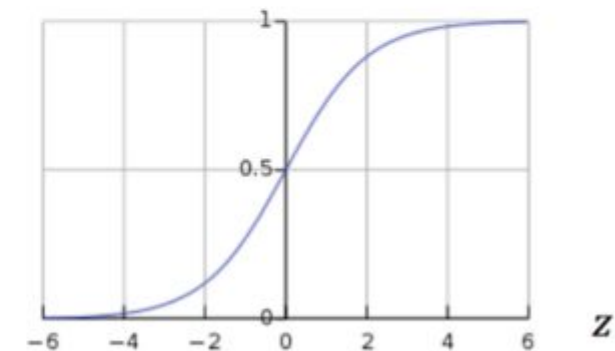
Neural Network in Machine Learning



$$\hat{y} = g(\theta_0 + \mathbf{X}^T \boldsymbol{\theta})$$

- Example: sigmoid function

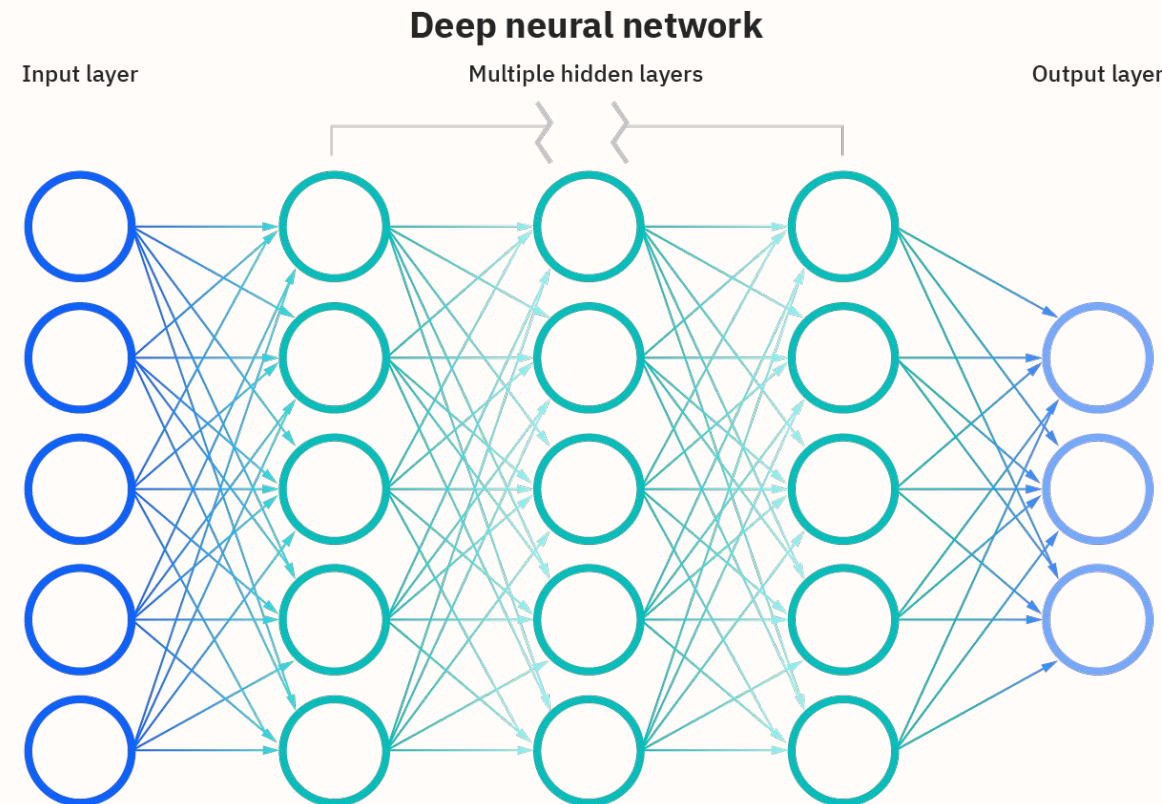
$$g(z) = \sigma(z) = \frac{1}{1 + e^{-z}}$$



2. Fundamental of Deep Learning

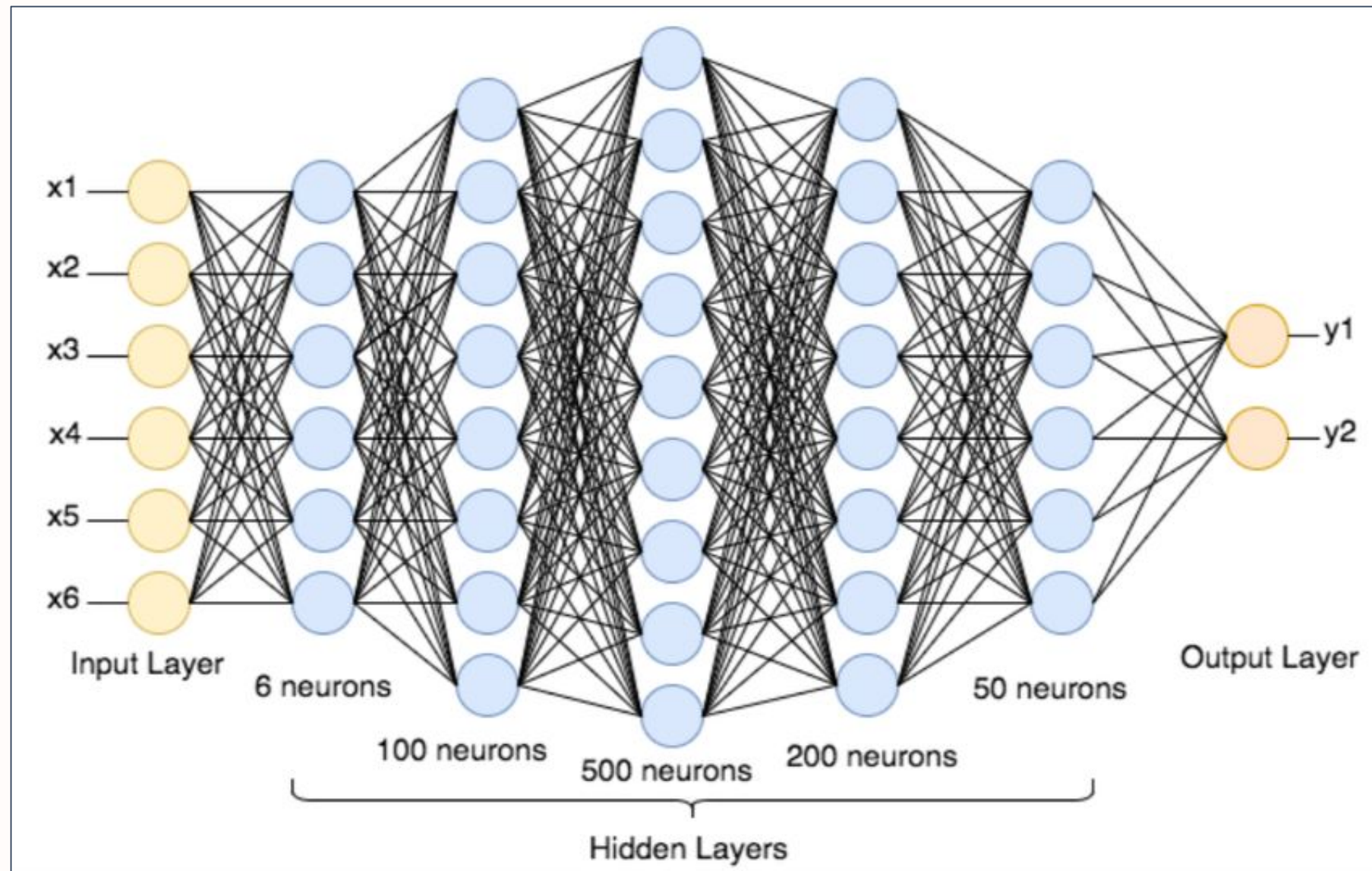
Definition

- **Deep learning** is a subset of machine learning that uses multi-layered neural networks, called **deep neural networks (DNN)**, to simulate the complex decision-making power of the human brain.
- By strict definition, DNN is a neural network with three or more layers. In practice, most DNNs have many more layers.



2. Fundamental of Deep Learning

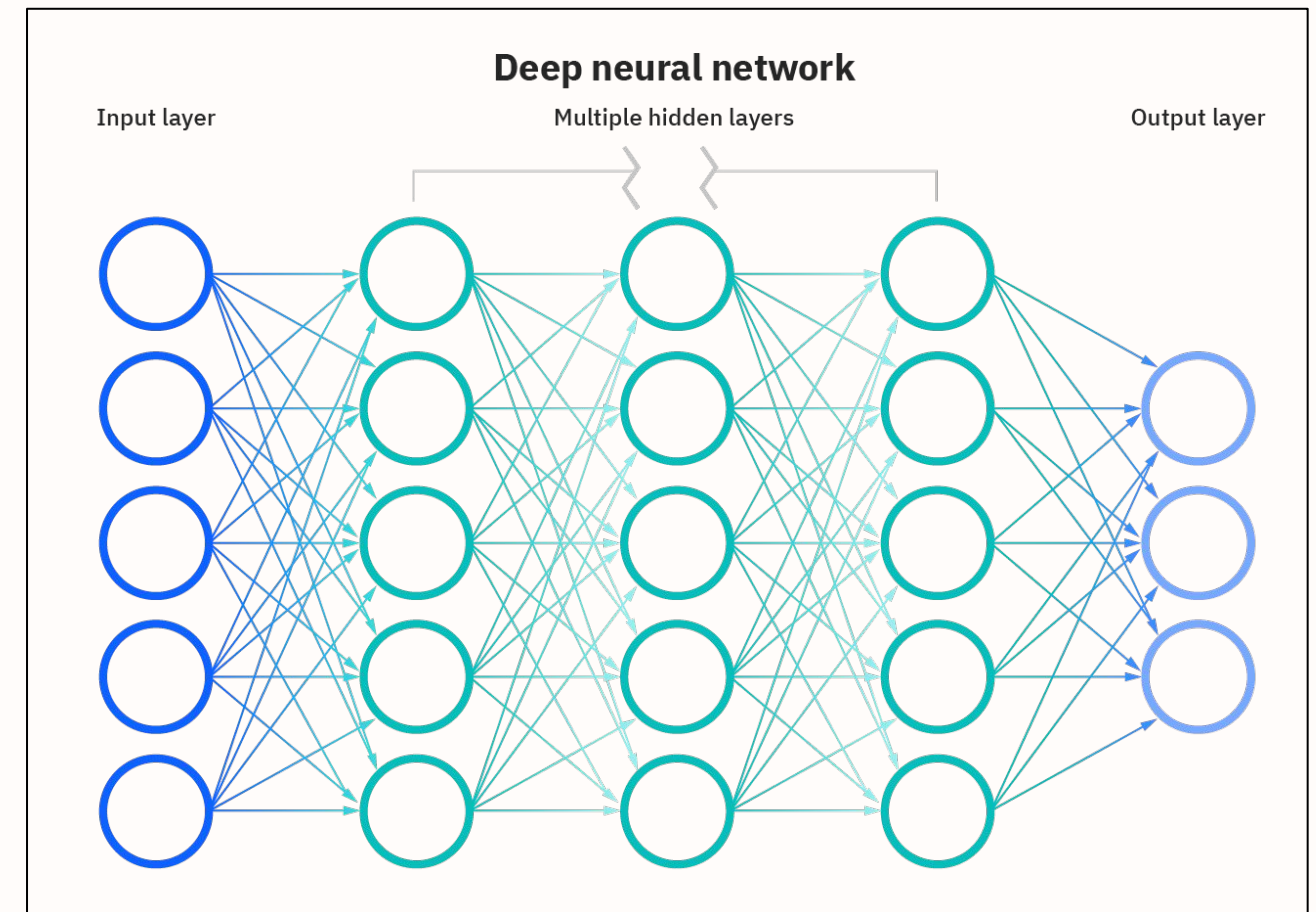
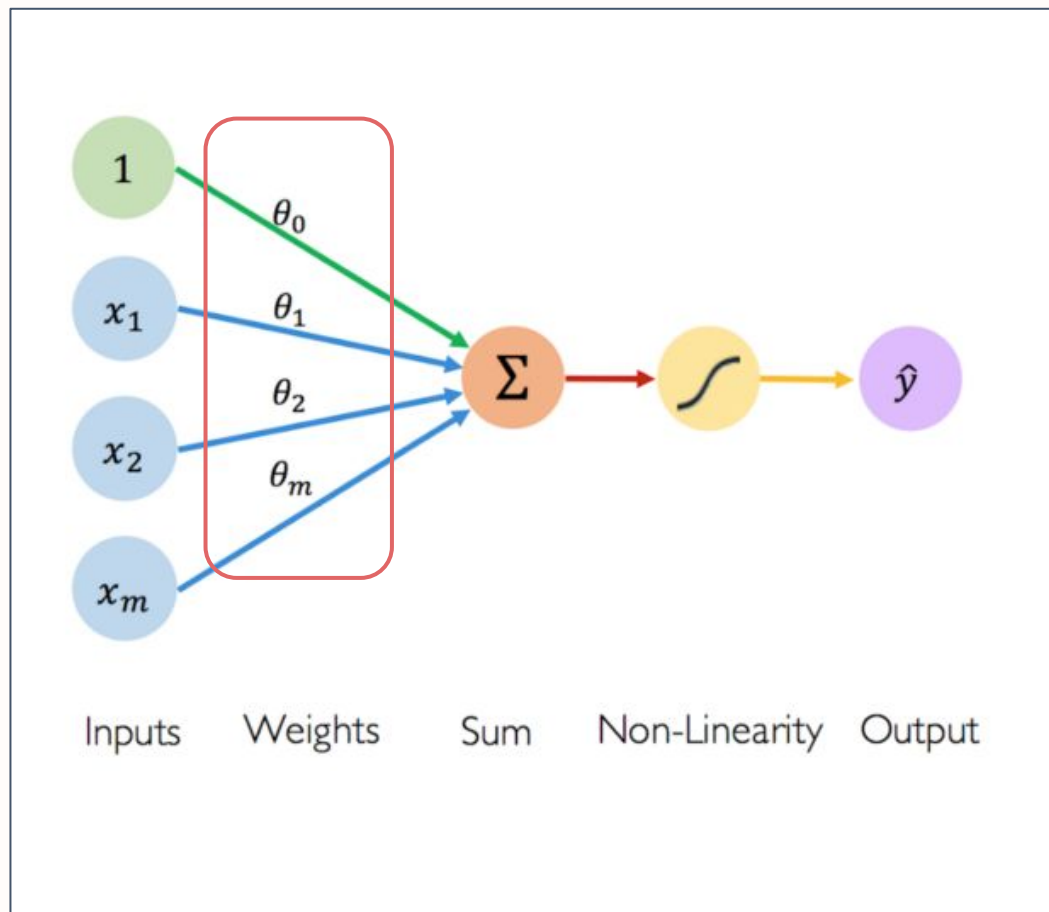
Definition



2. Fundamental of Deep Learning

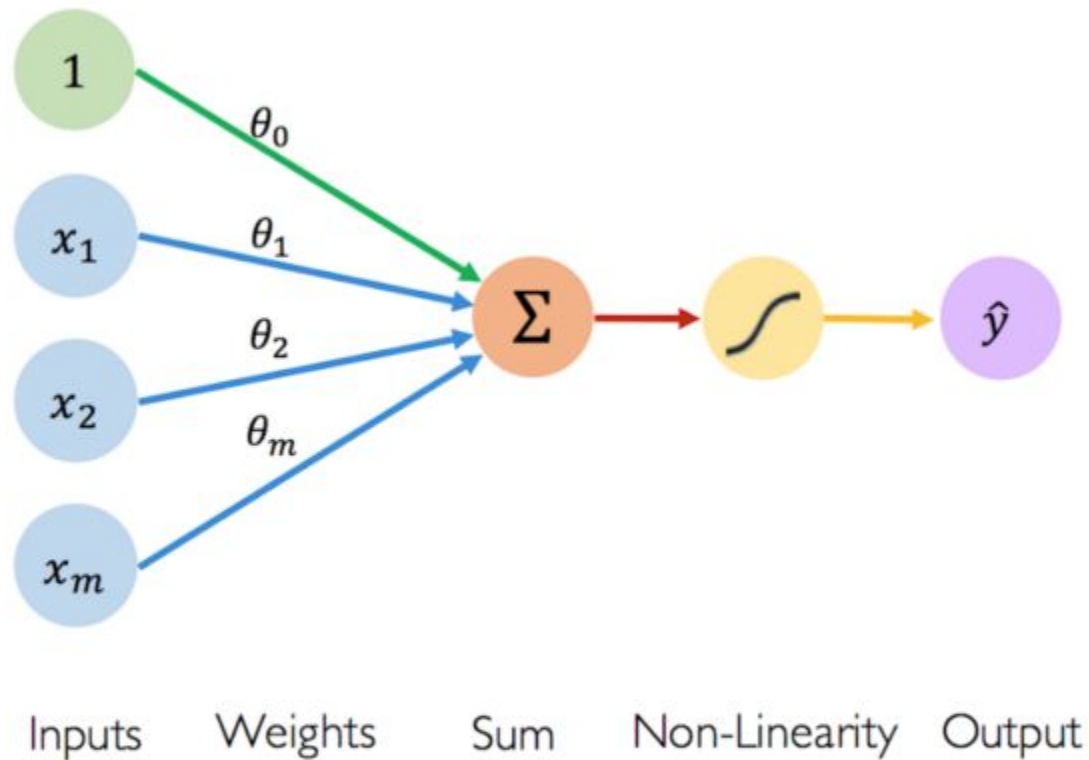
Weights

- **Weights** are numerical values associated with the connections between neurons. They determine the **strength of these connections** and, in turn, the influence that one neuron's output has on another neuron's input.



2. Fundamental of Deep Learning

Activation Function

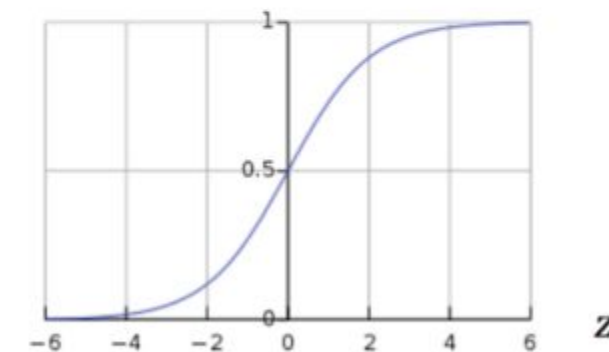


Activation Functions

$$\hat{y} = g(\theta_0 + \mathbf{X}^T \boldsymbol{\theta})$$

- Example: sigmoid function

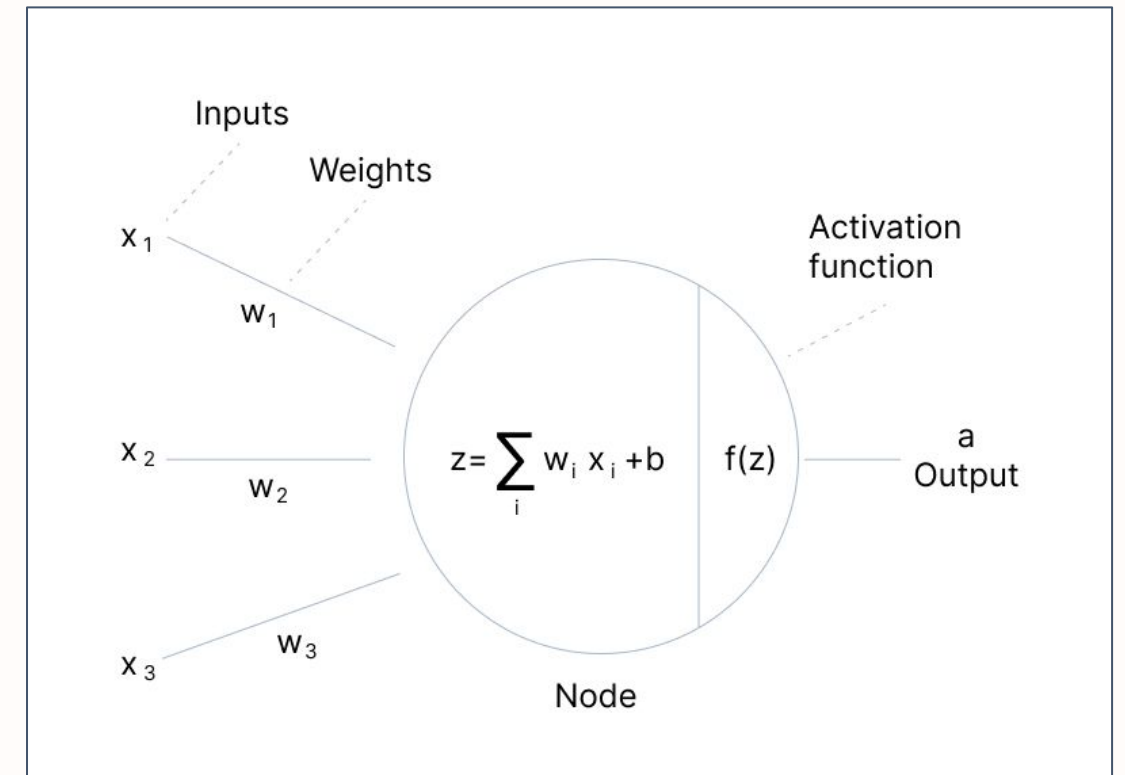
$$g(z) = \sigma(z) = \frac{1}{1 + e^{-z}}$$



2. Fundamental of Deep Learning

Activation Function

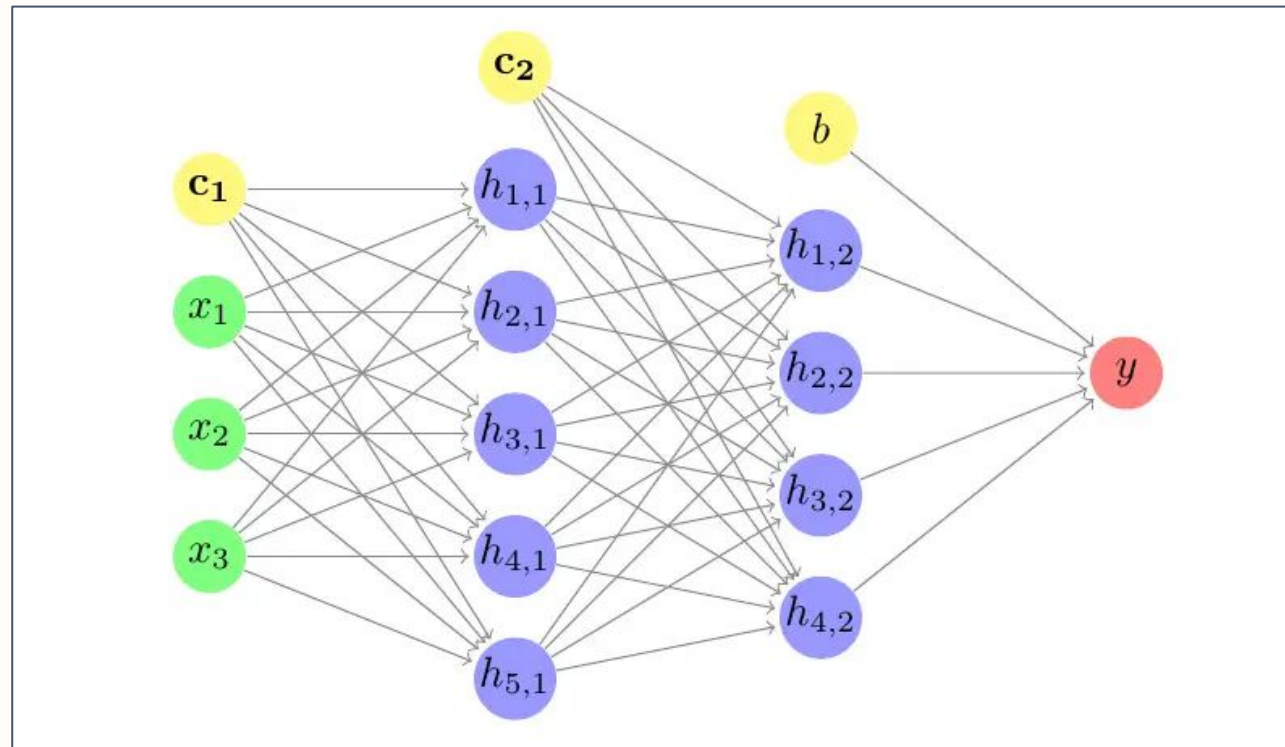
- The activation function decides *whether a neuron should be activated or not* by calculating the weighted sum and further adding bias to it.
- The purpose of the activation function is to *introduce non-linearity into the output* of a neuron.



2. Fundamental of Deep Learning

Bias

- Biases are essentially constants associated with each neuron. Unlike weights, biases are not connected to specific inputs but are added to the neuron's output.
- Biases allow neurons to activate even when the weighted sum of their inputs is not sufficient on its own.



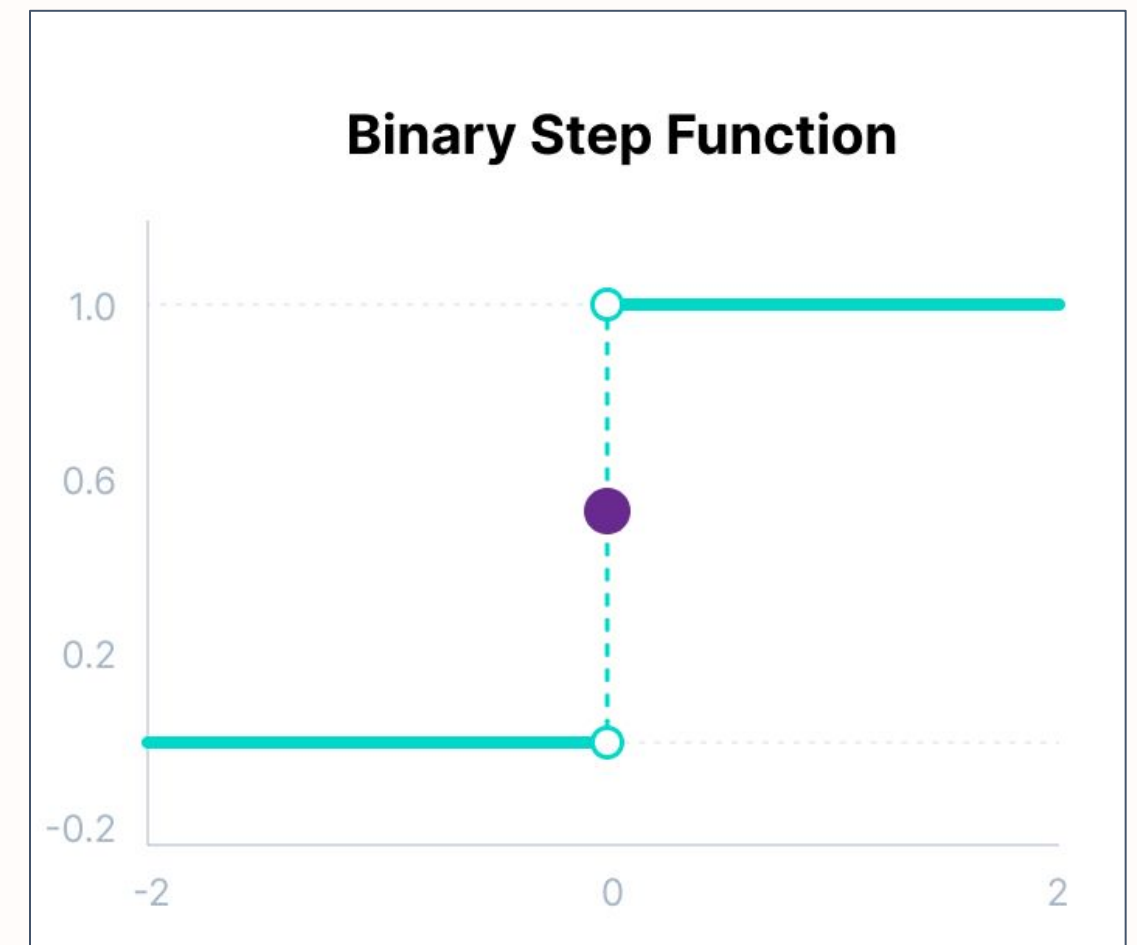
2. Fundamental of Deep Learning

Activation Function

a. Binary Step Function

$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$$

- It cannot provide multi-value outputs - for example, it cannot be used for multi-class classification problems.
- The gradient of the step function is zero.



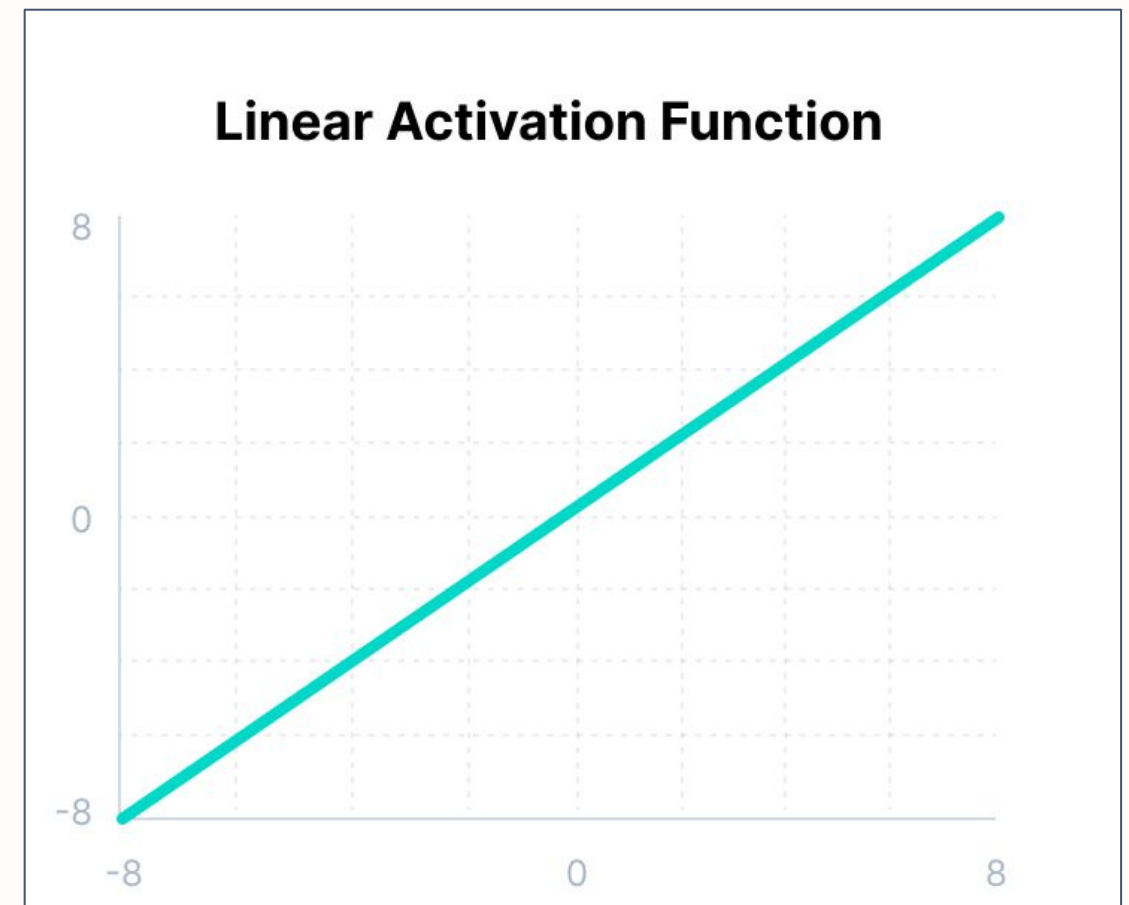
2. Fundamental of Deep Learning

Activation Function

b. Linear Function

$$f(x) = x$$

- The derivative of the function is a constant and has no relation to the input x.
- So, that the weights and biases will be updated during the optimization process but the updating factor would be the same.



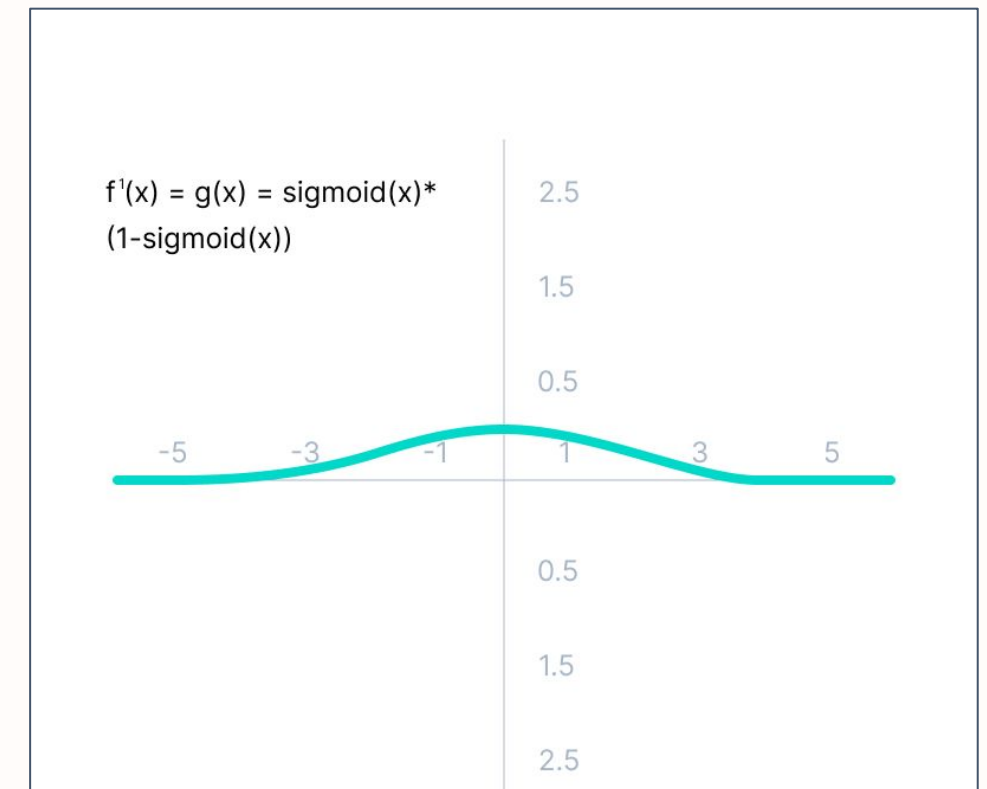
2. Fundamental of Deep Learning

Activation Function

c. Sigmoid Function

- The function is differentiable and provides a smooth gradient, i.e., preventing jumps in output values. This is represented by an S-shape of the sigmoid activation function.

$$f(x) = \frac{1}{1 + e^{-x}}$$

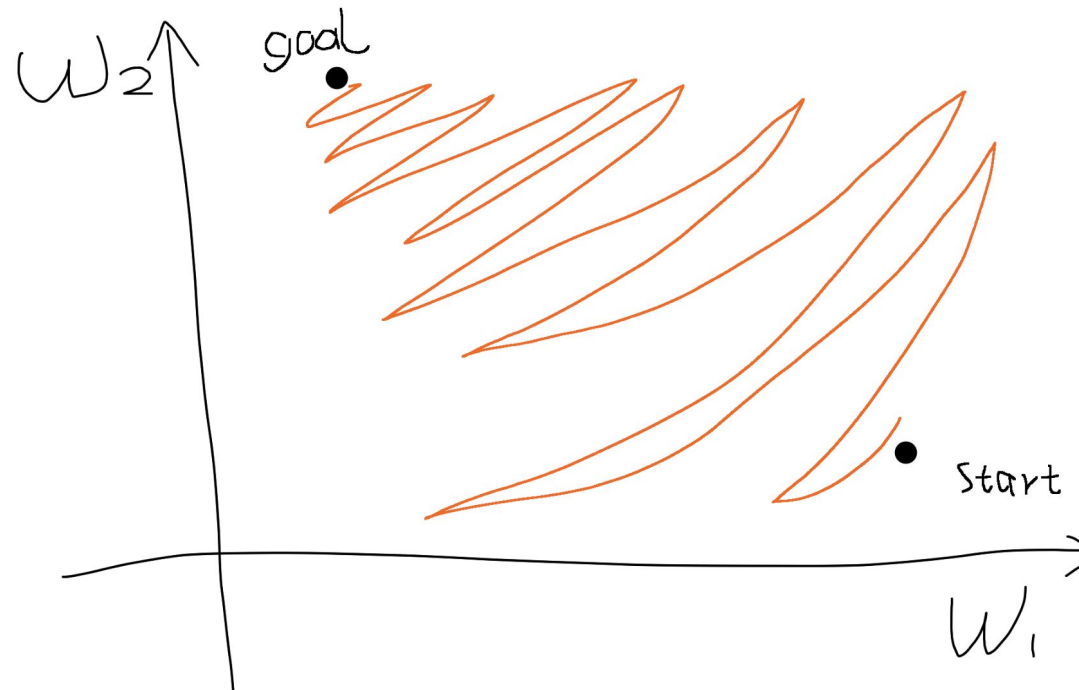


2. Fundamental of Deep Learning

Activation Function

c. Sigmoid Function

- This implies that for values greater than 3 or less than -3, will have very small gradients. As the gradient value approaches zero, the network is not really learning.
- Additionally, the sigmoid function is not symmetric around zero. So output of all the neurons will be of the same sign.



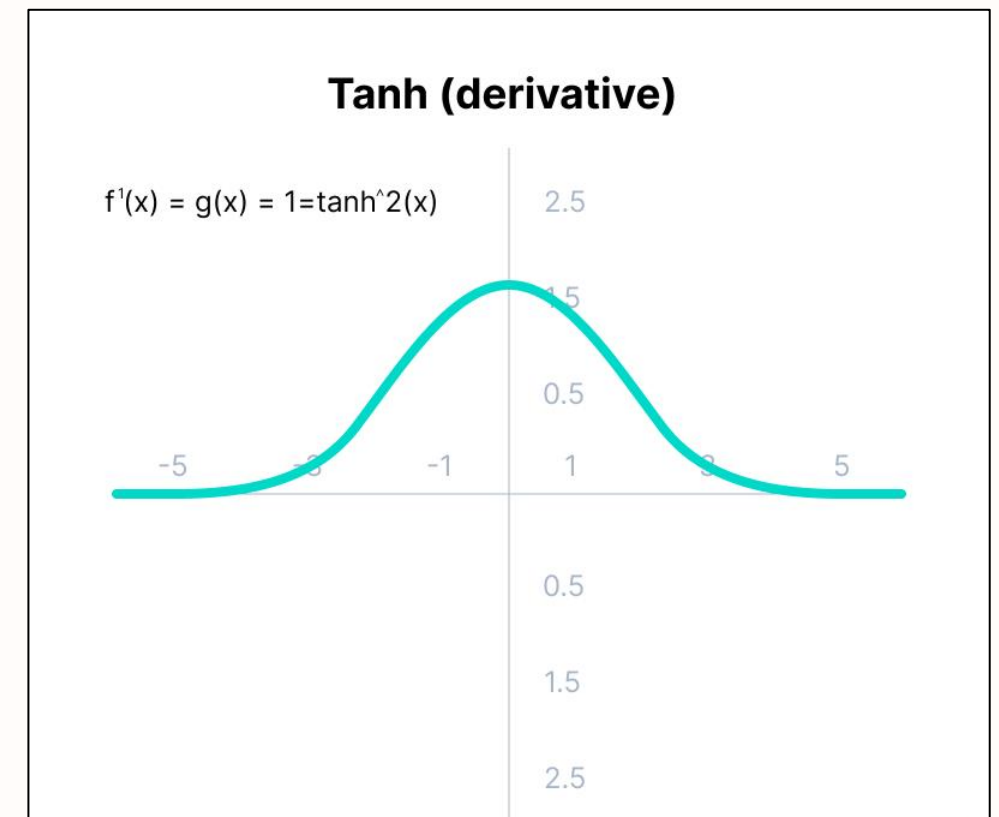
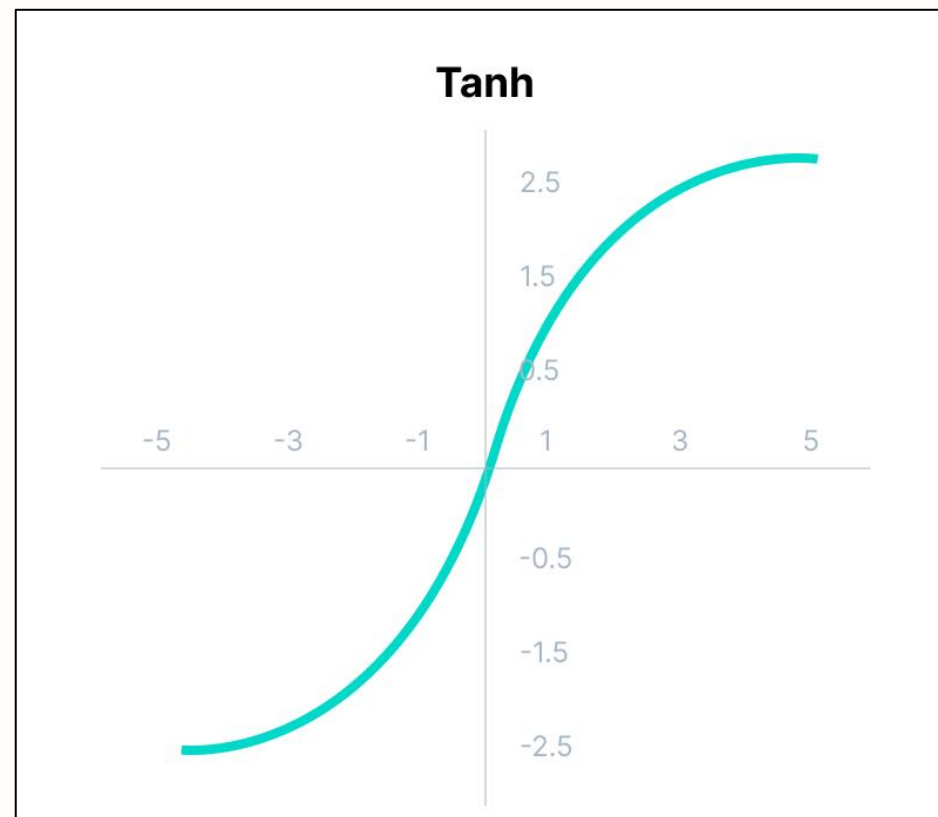
2. Fundamental of Deep Learning

Activation Function

c. Tanh Function

- The function is differentiable and provides a smooth gradient, i.e., preventing jumps in output values. This is represented by an S-shape of the sigmoid activation function.

$$f(x) = \frac{(e^x - e^{-x})}{(e^x + e^{-x})}$$

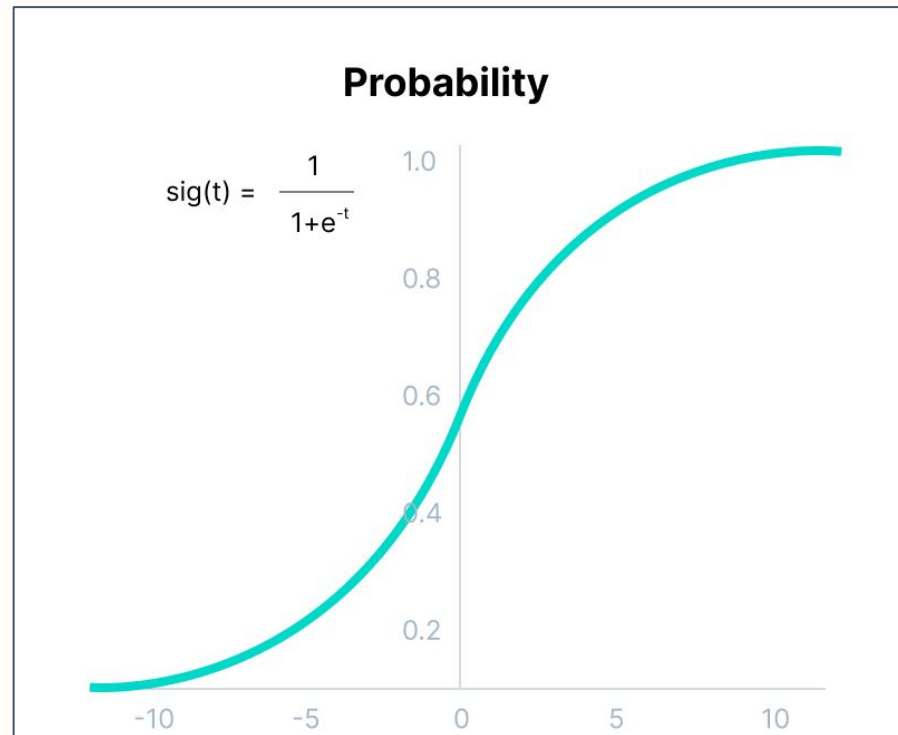


2. Fundamental of Deep Learning

Activation Function

d. Softmax Function

- The softmax function can be used for multiclass classification problems. This function returns the probability for a datapoint belonging to each individual class.
- Softmax function usually used for output layer rather than hidden layers.

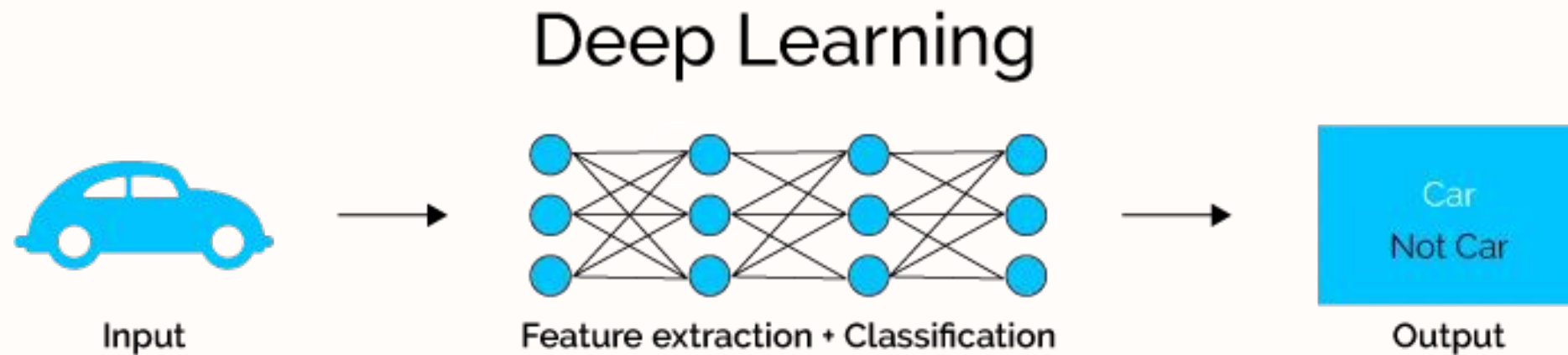
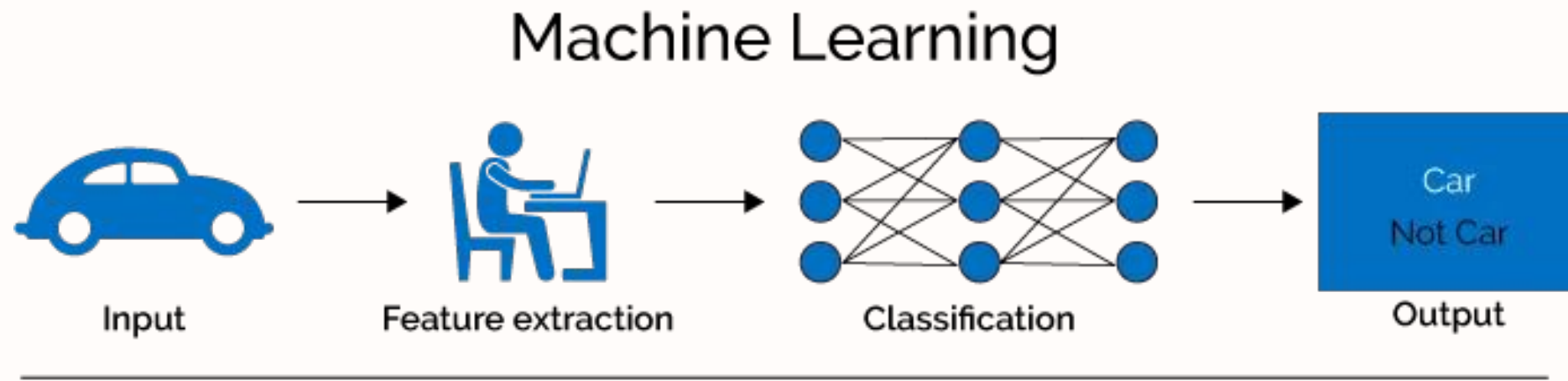


Softmax

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

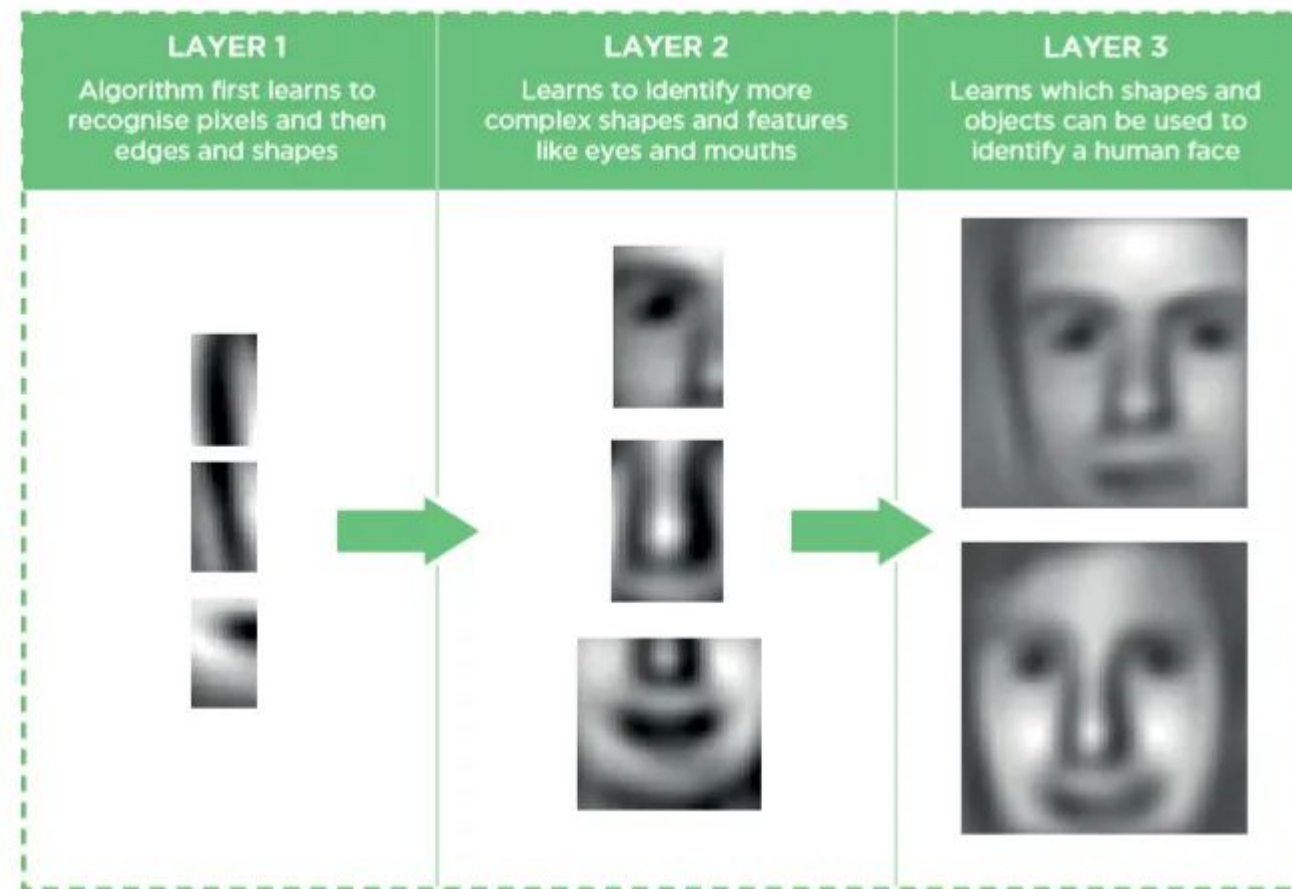
2. Fundamental of Deep Learning

Deep Learning vs. Machine Learning



2. Fundamental of Deep Learning

Deep Learning vs. Machine Learning



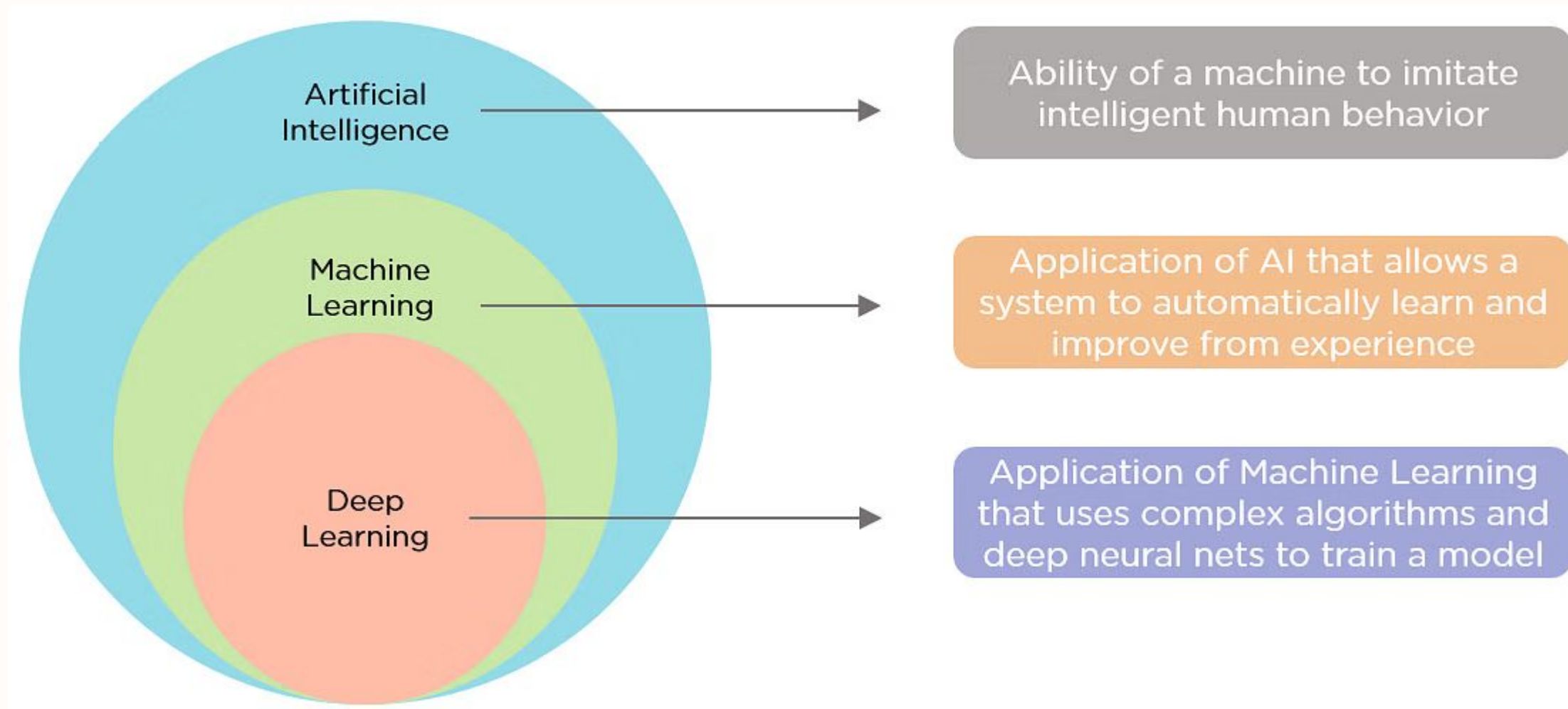
2. Fundamental of Deep Learning

Deep Learning vs. Machine Learning

Aspect	Machine Learning	Deep Learning
<i>Feature Extraction</i>	Manual feature engineering needed	Automatic feature extraction and representation learning
<i>Performance</i>	May have limitations on complex tasks	Excels in complex tasks, especially with big data
<i>Training Complexity</i>	Relatively simpler to train	Requires substantial computation power
<i>Algorithm Diversity</i>	Utilizes various algorithms like SVM, Random Forest	Mostly relies on neural networks
<i>Computational Demand</i>	Lighter computational requirements	Heavy computational demand
<i>Scalability</i>	May have limitations in scaling up	Scales well with increased data and resources

2. Overview of Deep Learning

Deep Learning vs. Machine Learning vs. Artificial Intelligent (AI)



3. Applications of DL

3.1. Computer Vision



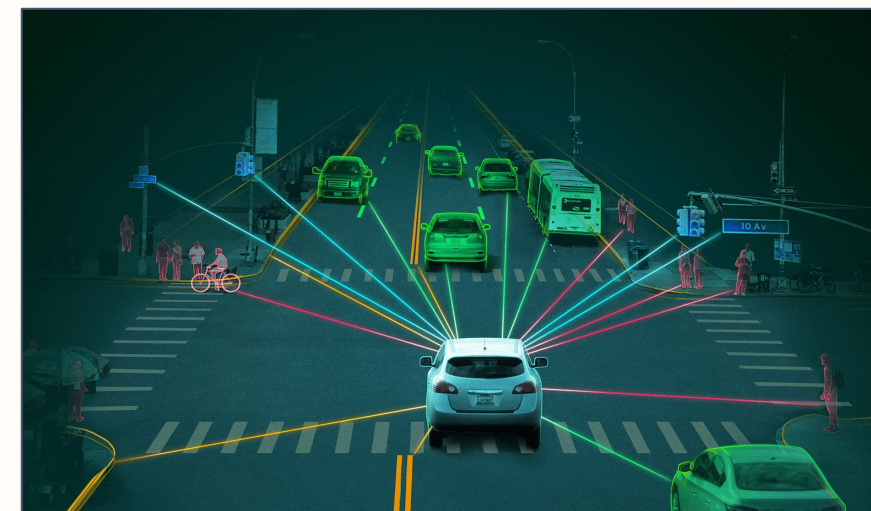
Object Detection
and Classification



Image
Segmentation



Face
Recognition



Autonomous
Driving

3. Applications of DL

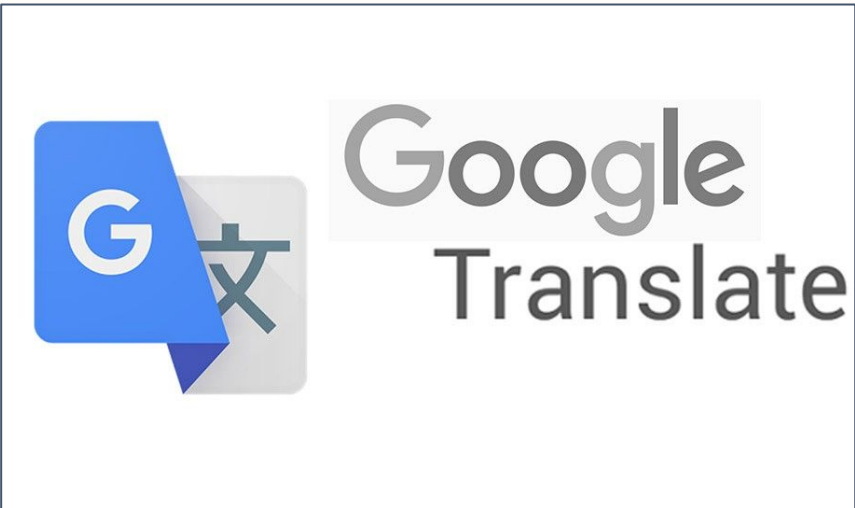
3.1. Computer Vision

Example: Autonomous Driving

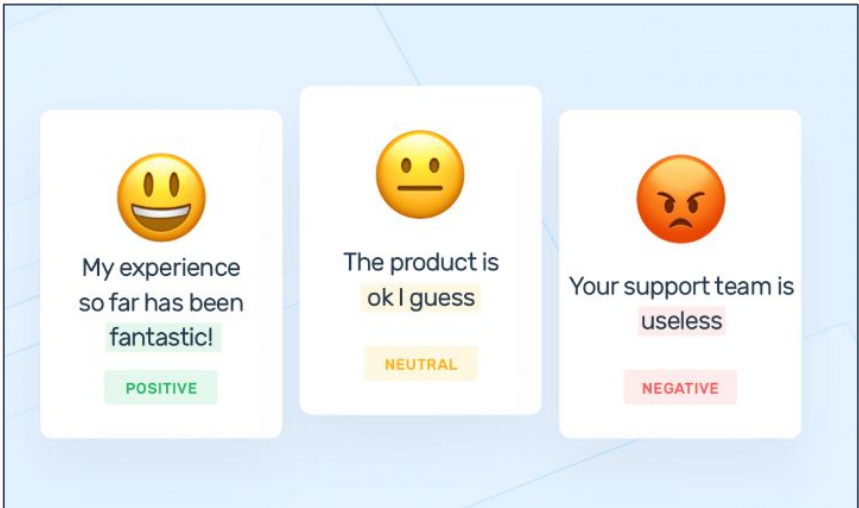
- Deep Learning in Camera Processing: Lane Detection, Traffic Light, Sign,...
- Deep Learning in Localization
- Deep Learning in Making Decision
- ...

3. Applications of DL

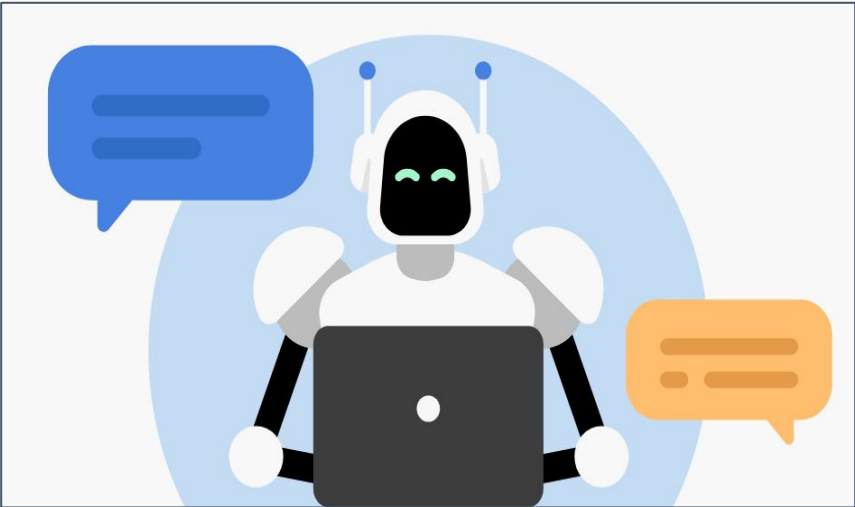
3.2. Natural Language Processing (NLP)



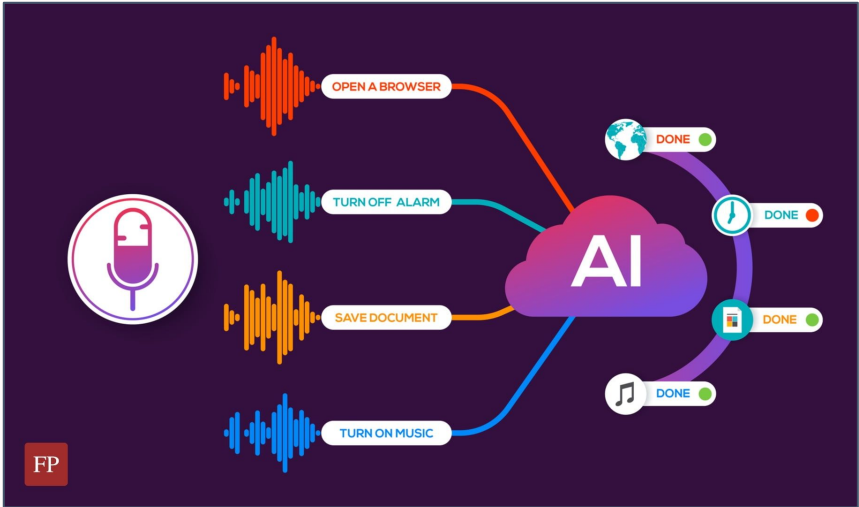
Language Translation



Sentiment Analysis



Text Generation



Speech Recognition

4. Limitations

- **Data Dependency:** Deep learning models **require large amounts of labeled data for training**, which may not always be available or easy to obtain, particularly in domains where data collection is expensive or time-consuming.
- **Computational Resources:** Training deep learning models can be **computationally intensive and time-consuming**, requiring specialized hardware such as GPUs (Graphics Processing Units) or TPUs (Tensor Processing Units).
- **Interpretability:** Deep learning models are often referred to as "**black boxes**" due to their complex architectures and millions of parameters, making it challenging to interpret how they arrive at their predictions.
- **Overfitting:** Overfitting occurs when a **model is too complex** and fits the training data too closely, causing it to perform poorly on new data.

5. Conclusion & Future Learning

Key terms of this section:

- Deep Learning operates on the principles of Neural Network.
- Deep Learning has many applications in the AI era.
- Deep Learning still has many limitations. One significant limitation is the "black box" problem.

In next section, we will:

- Dive into the intricacies of how a neural network learn from data.
- Explore the underlying mechanisms by which input information is transformed and propagated through the network's layers of interconnected neurons.
- Examine the role of activation functions, the flow of information through weighted connections, and the iterative process of learning and optimization that enables the network to make predictions or decisions based on the input data.

6. Q&A

